

SNAPDEAL

PRODUCT AND CATEGORY MANAGEMENT

1. CATEGORY TABLE

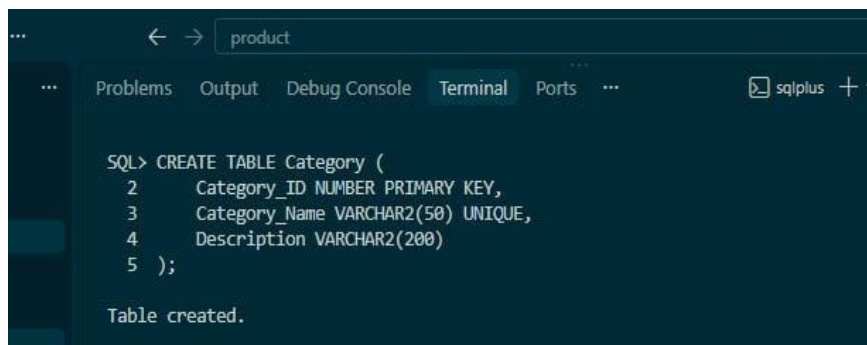
The Category table stores the different product categories in the Snapdeal database. It contains Category_ID, Category_Name, and Description.

Category	
Attribute	Key Type
Category_ID	Primary Key
Category_Name	Unique Key
Description	-

Categories: Footwear, Books, Beauty, Grocery, Sports

2. CREATE CATEGORY TABLE

The CREATE TABLE command is used to create the Category table. Category_ID is the Primary Key and Category_Name is Unique.

A screenshot of a SQL terminal window with a dark blue background. The terminal shows the command to create a table named 'Category'. The command is: SQL> CREATE TABLE Category (2 Category_ID NUMBER PRIMARY KEY, 3 Category_Name VARCHAR2(50) UNIQUE, 4 Description VARCHAR2(200) 5);. Below the command, the output 'Table created.' is displayed. The terminal window has tabs for 'Problems', 'Output', 'Debug Console', 'Terminal', and 'Ports'. The 'Terminal' tab is active. There is a search bar at the top with the text 'product' and a 'sqlplus' icon on the right.

3. Insert Data into Category

The INSERT command is used to add category records into the Category table.

CODE

INSERT INTO Category VALUES

(1, 'Footwear', 'Shoes and footwear products');

INSERT INTO Category VALUES

(2, 'Books', 'Educational and general books');

INSERT INTO Category VALUES

(3, 'Beauty', 'Beauty and personal care products');

INSERT INTO Category VALUES

(4, 'Grocery', 'Food and grocery products');

INSERT INTO Category VALUES

(5, 'Sports', 'Sports and fitness products');

COMMIT;

```
SQL> INSERT INTO Category VALUES
  2  (1, 'Footwear', 'Shoes and footwear products');

1 row created.

SQL>
SQL> INSERT INTO Category VALUES
  2  (2, 'Books', 'Educational and general books');

1 row created.

SQL>
SQL> INSERT INTO Category VALUES
  2  (3, 'Beauty', 'Beauty and personal care products');

1 row created.

SQL>
SQL> INSERT INTO Category VALUES
  2  (4, 'Grocery', 'Food and grocery products');

1 row created.

SQL>
SQL> INSERT INTO Category VALUES
```

4. Create Product Table

The CREATE TABLE command is used to create the Product table. Category_ID connects the Product table with the Category table.

```
SQL> CREATE TABLE Product (
  2    Product_ID NUMBER PRIMARY KEY,
  3    Product_Name VARCHAR2(100) NOT NULL,
  4    Category_ID NUMBER,
  5    Price NUMBER(10,2) NOT NULL,
  6    Stock NUMBER NOT NULL,
  7    Description VARCHAR2(200),
  8    FOREIGN KEY (Category_ID) REFERENCES Category(Category_ID)
  9 );

Table created.
```

5. Insert Data into Product

The INSERT command is used to add product records into the Product table.

CODE:

```
INSERT INTO Product VALUES  
(23, 'Nike Running Shoes', 1, 4999, 15,  
'Lightweight running shoes');
```

```
INSERT INTO Product VALUES  
(24, 'Adidas Casual Shoes', 1, 3599, 20,  
'Comfortable casual shoes');
```

```
INSERT INTO Product VALUES  
(25, 'Puma Sports Shoes', 1, 2999, 18,  
'Durable sports shoes');
```

```
INSERT INTO Product VALUES  
(26, 'Atomic Habits', 2, 499, 30,  
'Self improvement book');
```

```
INSERT INTO Product VALUES  
(27, 'Python Programming', 2, 699, 25,  
'Beginner programming book');
```

```
INSERT INTO Product VALUES  
(28, 'Rich Dad Poor Dad', 2, 399, 35,  
'Personal finance book');
```

```
INSERT INTO Product VALUES  
(29, 'Lakme Face Cream', 3, 399, 40,  
'Moisturizing face cream');
```

```
INSERT INTO Product VALUES  
(30, 'Maybelline Lipstick', 3, 599, 30,
```

'Long lasting lipstick');

INSERT INTO Product VALUES

(31, 'Nivea Body Lotion', 3, 349, 45,

'Moisturizing body lotion');

INSERT INTO Product VALUES

(32, 'Aashirvaad Atta 5kg', 4, 299, 50,

'Whole wheat flour');

INSERT INTO Product VALUES

(33, 'Tata Salt 1kg', 4, 30, 60,

'Refined iodized salt');

INSERT INTO Product VALUES

(34, 'Bru Instant Coffee 100g', 4, 275, 40,

'Instant coffee powder');

INSERT INTO Product VALUES

(35, 'Yoga Mat', 5, 799, 25,

'Non slip exercise yoga mat');

INSERT INTO Product VALUES

(36, 'Cosco Cricket Bat', 5, 1499, 15,

'Lightweight cricket bat');

INSERT INTO Product VALUES

(37, 'Nivia Football', 5, 999, 20

'Durable football for outdoor play');

COMMIT;

```

SQL> INSERT INTO Product VALUES
  2 (23, 'Nike Running Shoes', 1, 4999, 15,
  3 'Lightweight running shoes');

1 row created.

SQL>
SQL> INSERT INTO Product VALUES
  2 (24, 'Adidas Casual Shoes', 1, 3599, 20,
  3 'Comfortable casual shoes');

1 row created.

SQL>
SQL> INSERT INTO Product VALUES
  2 (25, 'Puma Sports Shoes', 1, 2999, 18,
  3 'Durable sports shoes');

1 row created.

```

6.Display Category Table

The SELECT command is used to display all records from the Category table.:

```
SELECT * FROM Category;
```

7.Display Product Table

The SELECT command is used to display all records from the Product table.

```

SQL> SELECT * FROM Product;

PRODUCT_ID
-----
PRODUCT_NAME
-----
CATEGORY_ID    PRICE    STOCK
-----
DESCRIPTION
-----
      23
Nike Running Shoes
      1      4999      15
Lightweight running shoes

PRODUCT_ID
-----
PRODUCT_NAME
-----
CATEGORY_ID    PRICE    STOCK
-----
DESCRIPTION
-----
      24
Adidas Casual Shoes
      1      3599      20
Comfortable casual shoes

PRODUCT_ID
-----
PRODUCT_NAME
-----

```

9.Update Product Stock

The UPDATE command is used to change the stock of Product_ID 32.

Code:

```
UPDATE Product
SET Stock = 30
WHERE Product_ID = 32;
```

10.Update Product Name and Price

The UPDATE command is used to change the product name and price of Product_ID 35.

Code:

```
UPDATE Product
SET Product_Name = 'Premium Yoga Mat',
    Price = 899
WHERE Product_ID = 35;
COMMIT;
```

11.Delete Product

The DELETE command is used to remove Product_ID 37 from the Product table.

Code:

```
DELETE FROM Product
WHERE Product_ID = 37;
COMMIT;
```

12.Join Category and Product

The JOIN operation combines the Category and Product tables using Category_ID.

```
SQL> SELECT
  2     c.Category_Name,
  3     p.Product_Name,
  4     p.Price,
  5     p.Stock
  6 FROM Category c
  7 JOIN Product p
  8 ON c.Category_ID = p.Category_ID
  9 ORDER BY c.Category_Name;
```

CATEGORY_NAME	PRODUCT_NAME	PRICE	STOCK
Beauty	Lakme Face Cream	399	40
Beauty	Nivea Body Lotion	349	45

13. Count Products by Category

The COUNT function is used to find the total number of products in each category.

```
SQL> SELECT
  2     c.Category_Name,
  3     COUNT(p.Product_ID) AS Total_Products
  4 FROM Category c
  5 LEFT JOIN Product p
  6 ON c.Category_ID = p.Category_ID
  7 GROUP BY c.Category_Name
  8 ORDER BY c.Category_Name;
```

CATEGORY_NAME	TOTAL_PRODUCTS
Beauty	3
Books	3
Footwear	3
Grocery	3
Sports	2

Conclusion :

The Snapdeal database was successfully created using Category and Product tables. SQL operations such as INSERT, SELECT, UPDATE, DELETE, JOIN, and COUNT were implemented successfully. The project demonstrates effective database management and the relationship between products and categories.